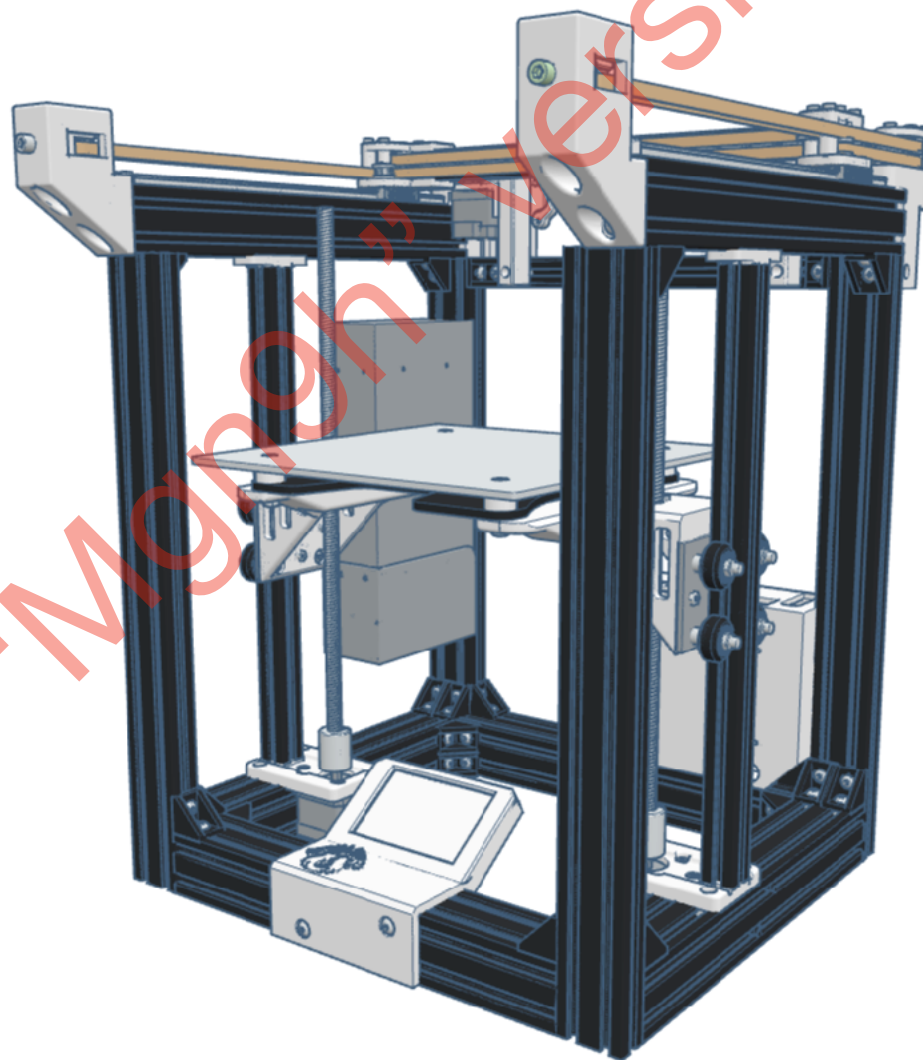




Duender
V-wheels AB & X axis



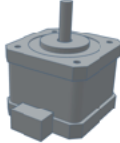
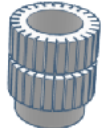
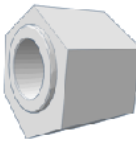
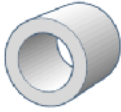
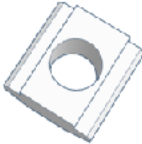
Your Open-Source 2 x Creality Ender-3 core XY conversion

by Sergei Irbis

Building the AB kinematics

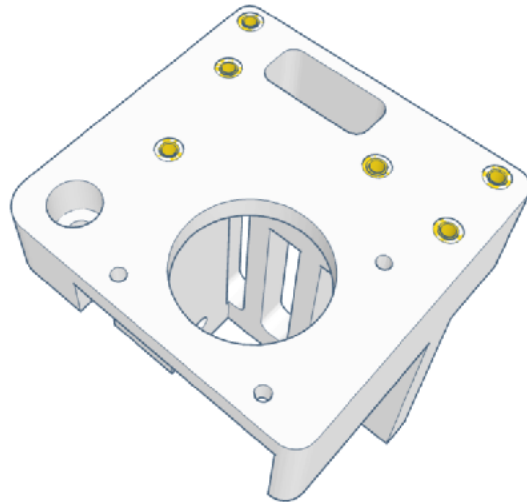
AB Gantry

22

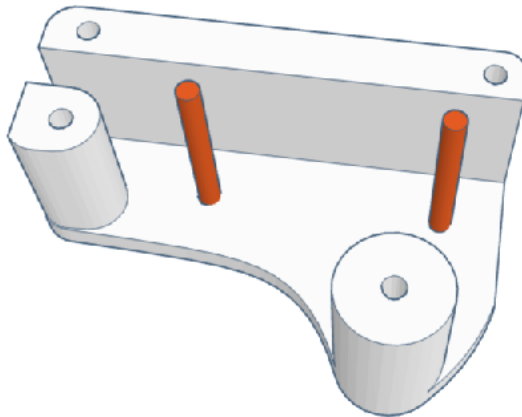
	Nema 17 42-40 (from Ender 3)	2		M3x10 spacer	1		M5x8 BHCS screw	14
	M3x30 SHCS	12		M3 Heat set insert	18		M5x30 BHCS screw (from Ender 3)	6
	M3x20 SHCS	2		GT2 16T6W3B toothless idler	8		Eccentric M5 nut (from Ender 3)	2
	M3x16 SHCS	2		GT2 16T6W5B motor pulley	2		Wheel M5 spacer (from Ender 3)	4
	M3x8 SHCS	4		M5 Nylstop nut (from Ender 3)	8		POM V-Wheel (from Ender 3)	6
	M3x0.5 spacer	28		M5 T-slot nut	2			

Building the AB kinematics

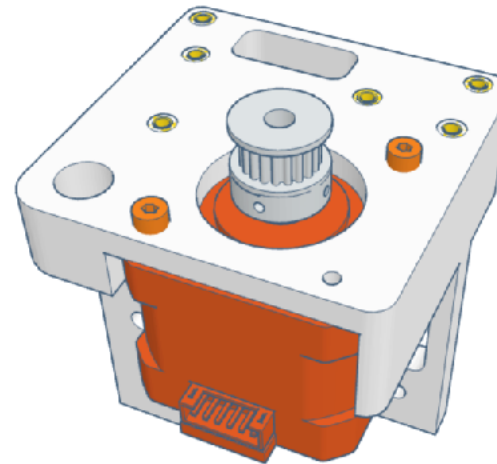
23



Prepare the bottom part of AB motor holders:
with 6 M3 heat-set inserts each side
(12 M3 heat-set inserts total).



Pass 2 M3x30 SHCS into the idler holes and turn both top halves upside down.



Install the motors under the plates with 2 M3x8 SHCS screws in the shown locations

Mind the motor plug orientation

Also place the motor pulleys onto the shafts of A & B motors

Do not tighten them too much. They will need to be adjusted later, after belts installation

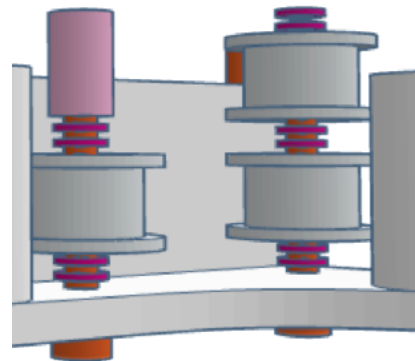
For the Left AB mount:

Onto the outermost pin (right side when upside down), stack:

- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers

On the innermost pin:

- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 more M3x0.5 mm spacers,
- 1 M3x10 mm spacer

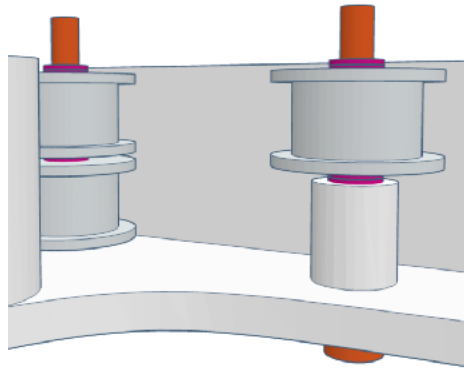


The best option is certainly to buy and install a metal M3x10 spacer, but should you not have one on hand, you may use the printed one provided in the stl files.

Building the AB kinematics

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For the Right AB mount:

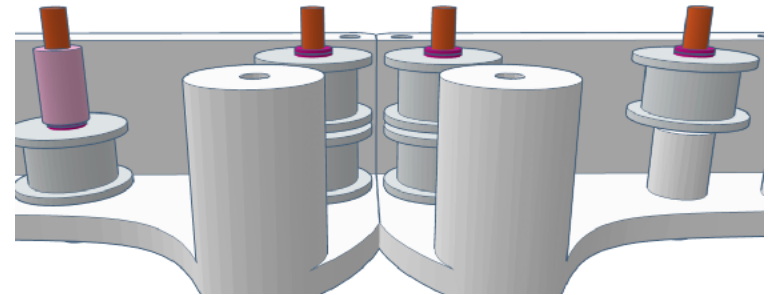


Onto the outermost pin (left side when upside down), stack:

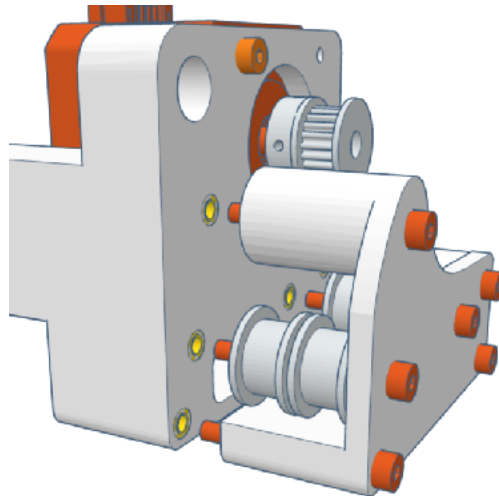
- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers.

Onto the innermost printed post pin:

- 3 M3x0.5 mm spacers,
- 1 GT2 16T toothless idler,
- 2 M3x0.5 mm spacers.



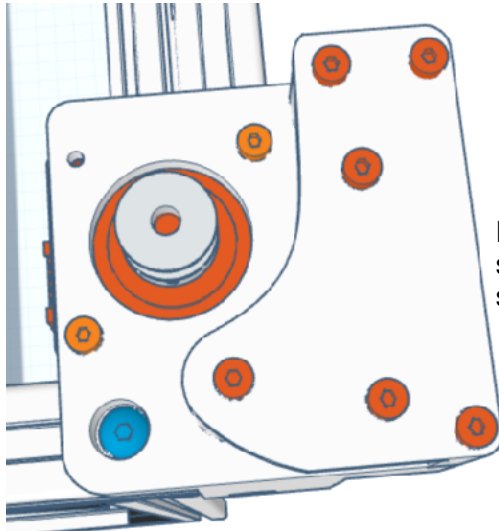
When the mounts are placed against each other, the idlers should be perfectly level between both sides. If that is not the case, adjust the spacer count as needed.



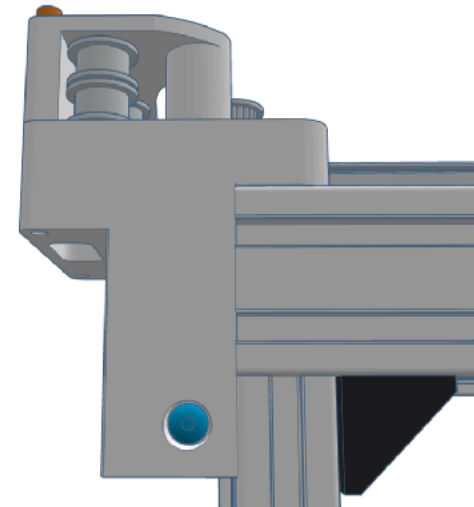
Lay the assembly on its back to prevent the spacers from falling out and assemble the two halves together.
Secure them adding 4 M3x30 SHCS on each side.
(12 M3x30 SHCS total)

Building the AB kinematics

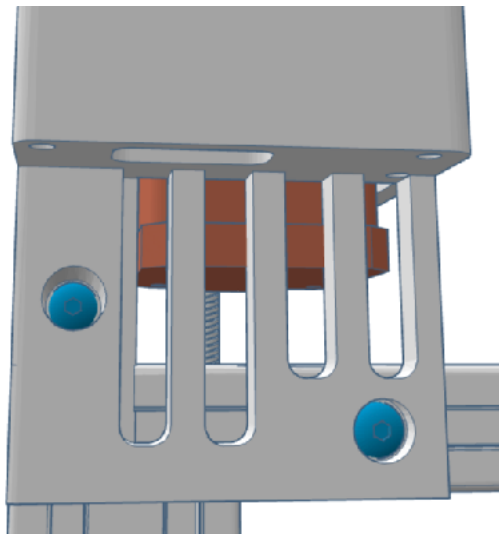
25



Install the AB motor holders on each side and use a M5x8 BHCS screw to secure it onto the top M5 T-slot nut



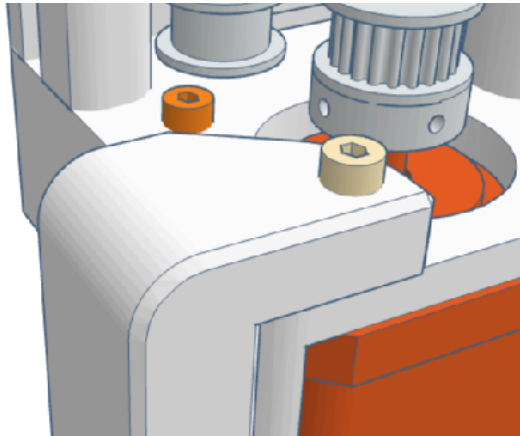
Use a second M5x8 BHCS secured into the tapped end of the 2020-E back brace



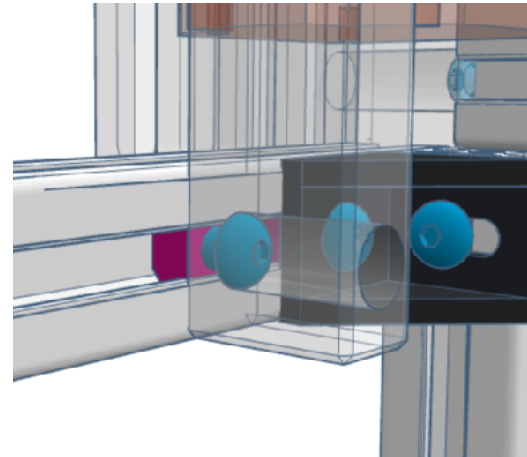
Use a M5x8 BHCS to hold into the tapped end of the 2040-C Y axis and another one to hold onto the back brace M5 T-slot nut

Building the AB kinematics

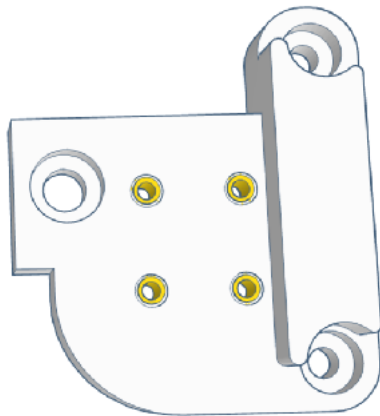
26



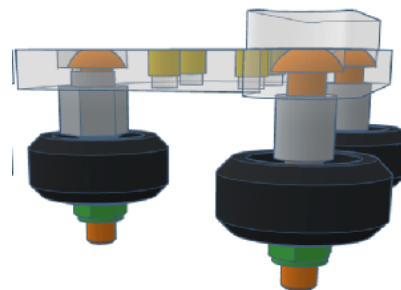
Install the motor holders with M3x16 SHCS threaded into the motor



Secure the base of the holder with an M5x8 BHCS screw into one of the previously inserted T-slot nuts in front of the back brace



Prepare both carriage plates by inserting 4 M3 heat-set inserts each side (8 M3 heat inserts total).



Just like the Z plates, insert 3 M5x30 BHCS into each carriage, then on the outer side, each takes:

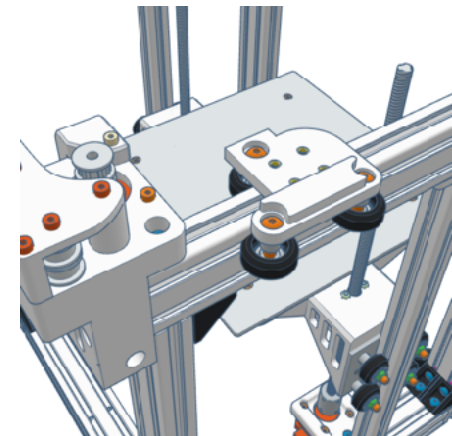
- 1 M5 spacer,
- 1 V-wheel,
- 1 M5 hex nut.

On the inner side, stack:

- 1 eccentric nut,
- 1 V-wheel,
- 1 M5 hex nut to tighten the whole.

Leave them loose for now.

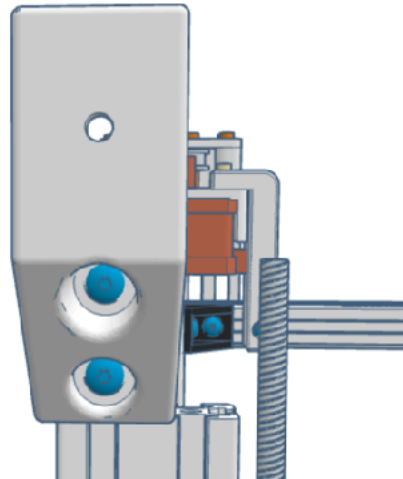
(6 M5x30 BHCS, 4 M5 spacers, 2 eccentric columns, 6 V-wheels and 6 M5 hex nuts total)



Place the carriages on both Y axis rails and tighten the M5 hex nuts, then adjust the eccentric nuts to set the carriage preload.

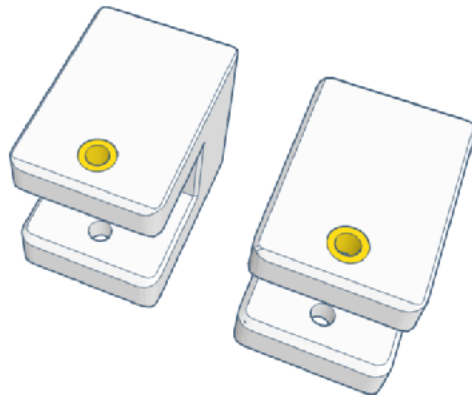
Building the AB kinematics

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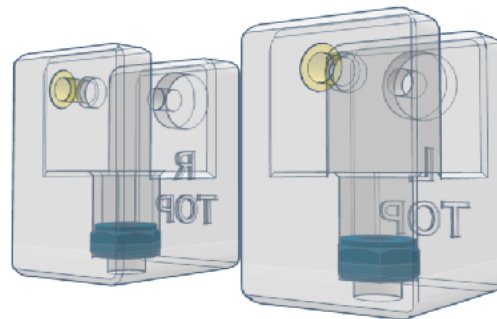


Install the front holders with 2 M5x8 BHCS on each side

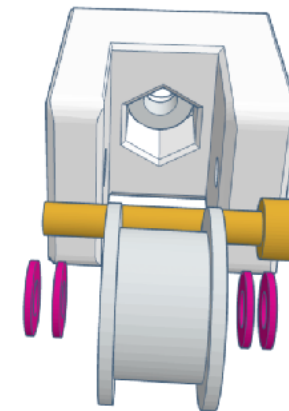
Mind the side, the holders are not symmetrical



Prepare both the left and right belt tensioners by placing an M3 heat-set insert in the underside hole



Insert an M5 Nylstop nut into the groove inside both holders. A small dab of glue may be used to keep them in place.



Place

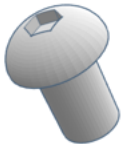
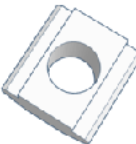
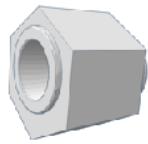
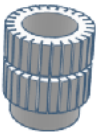
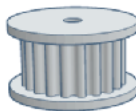
- 2 M3x0.5 mm spacers,
 - 1 16T toothless idler,
 - 2 M3x0.5 mm spacers,
- all held by a M3x20 SHCS screw inside each tensioner

As already mentioned, [a printed jig](#) can make this much easier

Building the X axis

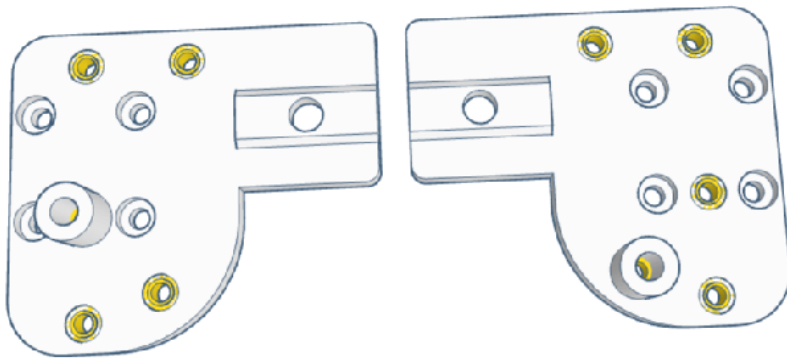
X axis gantry

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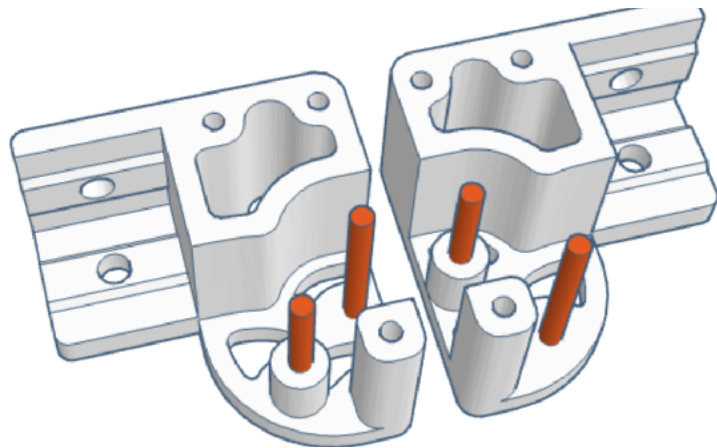
	M3x30 SHCS	10		M5x8 BHCS screw	6		M5x30 BHCS screw (from Ender 3)	3
	M3x6 SHCS	8		M5 T-slot nut	6		Eccentric M5 nut (from Ender 3)	1
	M3x0.5 spacer	20		GT216T6W3B toothless idler	2		Wheel M5 spacer (from Ender 3)	2
	M3 Heat set insert	10		GT2 16T6W3B toothed idler	2		POM V-Wheel (from Ender 3)	3
				Modified toolhead plate (from Ender 3)	1		M5 Nylstop nut (from Ender 3)	3

Building the X axis

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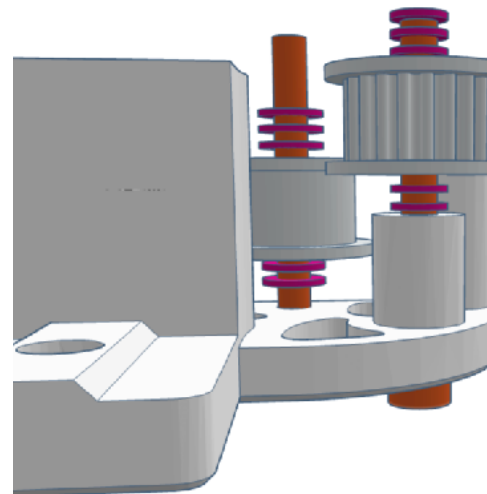
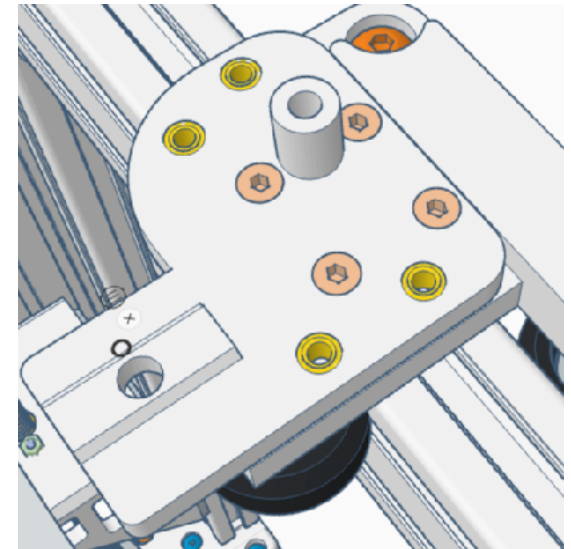


Prepare both bottom parts of the X carriage with 5 M3 heat-set inserts on each side.
Those in the printed posts are easier to insert from below.



Turn both top halves upside down, and insert 2 M3x30 SHCS each side to serve as the idler axis

Place each bottom half onto its carriage, and secure it with 4 M3x6 SHCS (8 M3x6 SHCS total).



Each half takes:

On the front axis

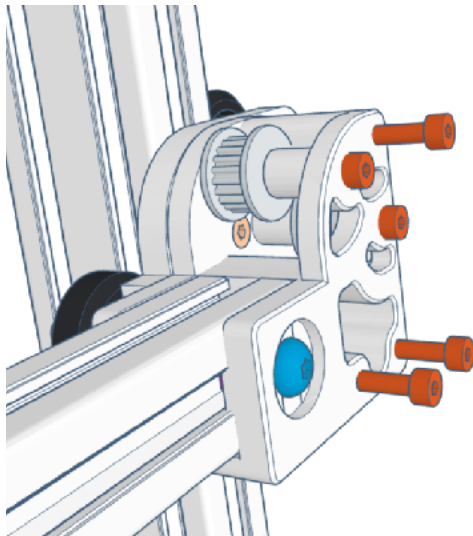
- 2 M3x0.5 mm spacers,
- 1 GT2 16T **toothed** idler,
- 3 M3x0.5 mm spacers

On the rear axis

- 2 M3x0.5 mm spacers,
- 1 GT2 16T **toothless** idler,
- 3 M3x0.5 mm spacers.

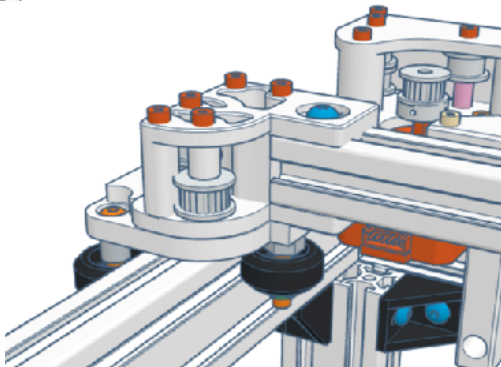
Building the X axis

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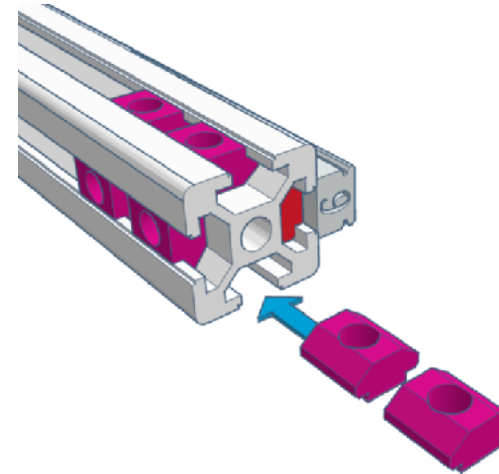
Lay the printer on its back to prevent the spacers from falling out and tighten both sides of the X-axis assembly, adding 3 M3x30 SHCS, to assemble the two halves.
(6 M3x30 SHCS total)

Just screw enough so that parts hold together: both halves need to be loose for the next step.

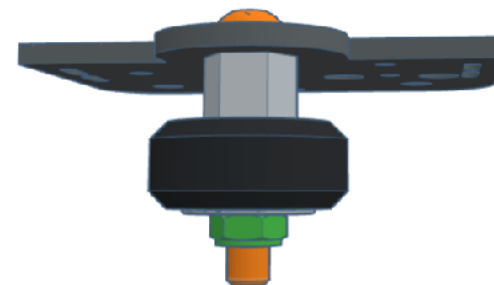


Push the X axis into its carriages on the two sides of the printer, carefully centre it, then tighten both carriages.

Once done, use the previously inserted T-nuts to tighten 3 M5x8 BHCS at each end:
1 on top, 1 on the back and 1 under the X axis.
(6 M5x8 BHCS total)



Insert 2 M5 T-slot nuts on the top side, 2 on the back side and 2 more under the X axis extrusion
(6 M5 T-slot nuts total)



Place a M5x30 BHCS into the bottom of the toolhead plate, from below, stack:

- 1 eccentric nut,
- 1 V-wheel,
- 1 M5 hex nut.

Leave the nut loose for now.

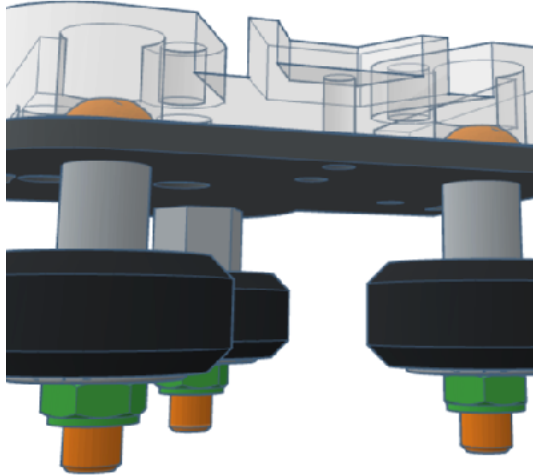
Onto each M5x30 BHCS through the top of the plate, stack:

- 1 wheel spacer,
- 1 V-wheel,
- 1 M5 hex nut.

Leave the nut loose for now.

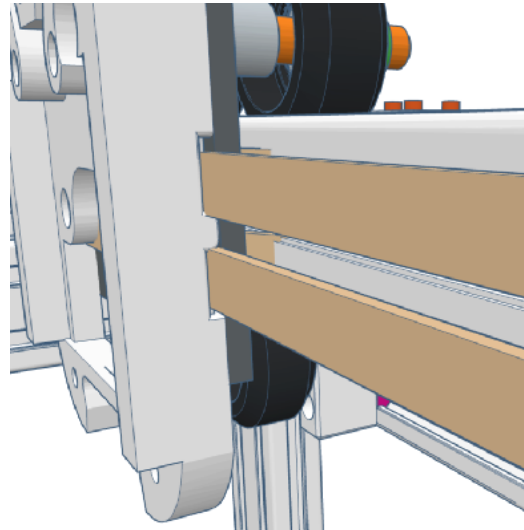
Building the X axis

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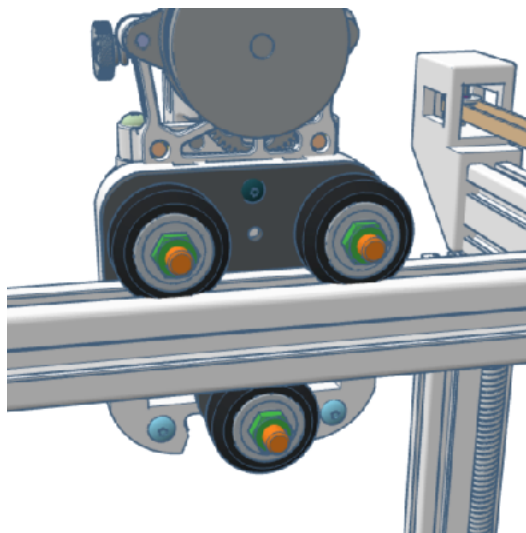
Put the adapter plate onto the metal carriage, and attach it using 2 M5x30 BHCS that hold a M5 spacer, a V-wheel and a M5 hex nut from below.

Leave the nuts loose for now.



Position the wheels onto the X extrusion, the belts (installed in next section) will pass through the side openings, be folded onto themselves behind the metal plate secured with a zip tie.

Once aligned, tighten the M5 nuts to secure the wheel assembly onto the extrusion, then adjust the eccentric nut to set the carriage preload.



Once assembled, the toolhead will be attached to the carriage using 2 M3x8 SHCS on the bottom, and 1 M3x18 SHCS on the top.